

1 Introduction

A generating function is a device somewhat similar to a bag. Instead of carrying many little objects detachedly, which could be embarrassing, we put them all in a bag, and then we have only one object to carry, the bag.

—George Pólya, *Mathematics and Plausible Reasoning*

A generating function is a clothesline on which we hang up a sequence of numbers for display.

—Herbert Wilf, *Generatingfunctionology*

Example 1.1 (Binet's formula). Let $F_0 = 0$, $F_1 = 1$, and $F_n = F_{n-1} + F_{n-2}$ for $n \geq 2$. Let $\phi = \frac{1+\sqrt{5}}{2}$ and $\psi = \frac{1-\sqrt{5}}{2}$. Show (using generating functions) that

$$F_n = \frac{\phi^n - \psi^n}{\sqrt{5}}.$$

Example 1.2 (Classic, suggested by Catherine Xu). Is it possible to re-label the faces of a pair of six-sided dice with positive integers such that the probability of rolling each sum from 2 to 12 stays the same?

2 Roots of unity filter

My genfunc skills are through the ROUF!

—my solution to a problem on MOP test 3

Example 2.1 (Classic). For a subset $S \subseteq \{1, 2, \dots, 2000\}$, let $\sigma(S)$ be the sum of its elements. How many $S \subseteq \{1, 2, \dots, 2000\}$ satisfy $5 \mid \sigma(S)$?

3 Problems

Problem 3.1 (Classic). Compute

$$\binom{100}{0} + \binom{100}{3} + \binom{100}{6} + \cdots + \binom{100}{99}.$$

Problem 3.2 (Classic). Compute

$$0 \binom{100}{0} + 1 \binom{100}{1} + 2 \binom{100}{2} + \cdots + 100 \binom{100}{100}.$$

Problem 3.3 (AIME II 2016/12). Place a regular hexagon in the plane. Compute the number of ways to paint each vertex one of four colors such that no two adjacent vertices can be the same color.

Problem 3.4 (HMMT Combo 2007/9). Let S denote the set of triples (i, j, k) of positive integers where $i + j + k = 17$. Compute

$$\sum_{(i,j,k) \in S} ijk.$$

Problem 3.5 (Project Euler 956). Let $\Omega(n)$ be the number of prime factors of n , counted with multiplicity. Let $D(n, m)$ be the sum of all divisors d of n where $\Omega(d)$ is divisible by m . Describe an efficient algorithm to compute

$$D\left(\prod_{i=1}^{1000} \prod_{j=1}^i j, 1000\right)$$

modulo the prime $10^9 - 999$.

Problem 3.6 (Putnam 2018 B6). Let S be the set of sequences of length 2018 whose terms are in the set $\{1, 2, 3, 4, 5, 6, 10\}$ and sum to 3860. Prove that the cardinality of S is at most

$$2^{3860} \cdot \left(\frac{2018}{2048}\right)^{2018}.$$

Problem 3.7 (Brian Reinhart from G2, suggested by Catherine Xu). A 4×4 grid containing squares labeled $1, 2, \dots, 16$ each exactly once is called *sensible* if, over all sets S of four squares containing exactly one square in each row and exactly one square in each column, the sum of the elements of S is constant. Compute the number of sensible grids.

Problem 3.8 (Kraft–McMillan from MATH 5251). For any set S , let S^* be the set of finite sequences of elements of S . The elements of S^* are called the *words* on S . For example, $1011 \in \{0, 1\}^*$.

Let $W = \{1, 2, \dots, m\}$ and $\Sigma = \{1, 2, \dots, n\}$ be finite sets. A *code* is a function f from W to Σ^* . Let f^* be the map defined by $f^*(w_1 w_2 \dots w_k) = f(w_1) f(w_2) \dots f(w_k)$, where we have concatenation on the right side. We say f is *uniquely decipherable* if f^* is injective.

Show that if f is uniquely decipherable, then

$$\sum_{i=1}^m \frac{1}{n^{|f(i)|}} \leq 1.$$